AWS Guidance Paper for
Dynamic Pricing Transformation in Retail Industry

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Introduction

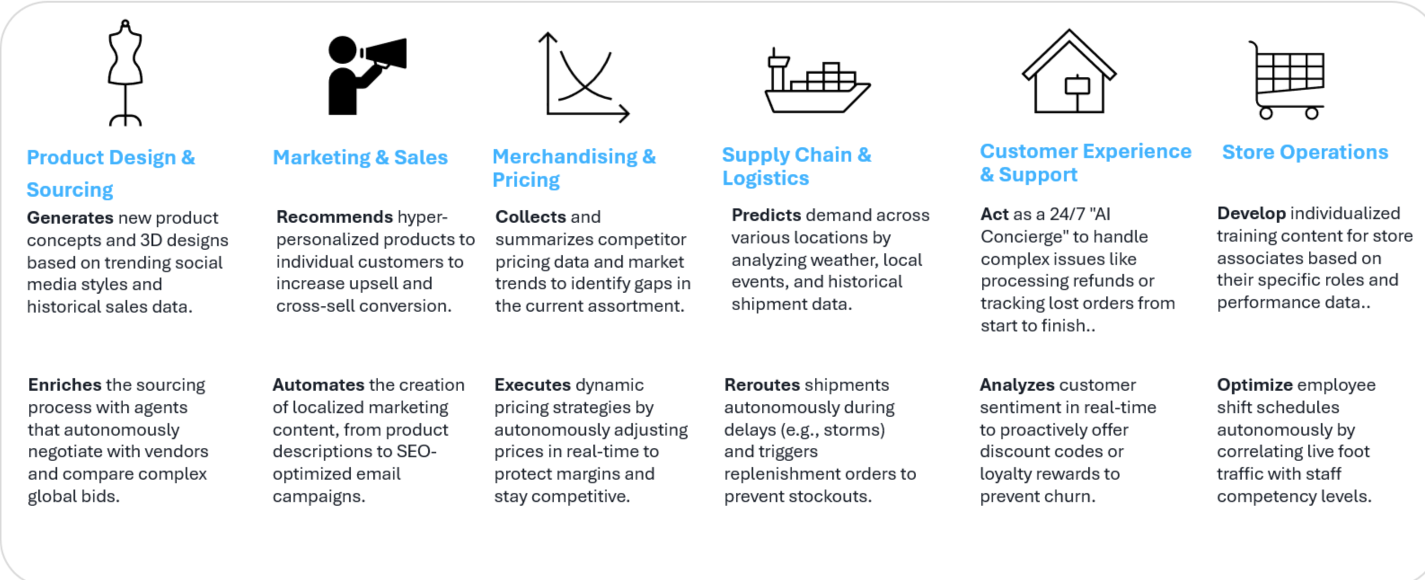
The Retail industry is undergoing a fundamental transformation driven by Agentic AI and GenAI across its entire value chain. It spans from sourcing and procurement through merchandising, inventory management, pricing, marketing, sales, and customer service. AI is fundamentally changing how customers shop, how retailers operate, and where profit flows, requiring retailers to redesign their customer value propositions, economics, capabilities, and operating models end to end.

Retail is shifting from search-based shopping to **agentic commerce**, where AI assistants guide shoppers from discovery to checkout in no time. This shift has redefined **Revenue Growth Management (RGM)**. Instead of looking at past months' data, systems now use live market signals and social trends to adjust pricing and inventory before a stock out happens or trend even peaks. Marketing has finally reached a level of **hyper-personalization**, using actual behavior to build trust through helpful interactions rather than generic ads. This agility is no longer just an insight on a dashboard; AI has become an autonomous operator embedded across the business operations and value chain. It acts as an intelligent resource that can assist store associates or independently reroute supply chains during a crisis. Ultimately, the industry is moving toward a responsive, autonomous network that fulfills a consumer's needs before they even realize they have them.

This document details how the **AWS Agentic AI stack** transforms retail pricing from a manual, 6–10-week process into an autonomous system delivering decisions in hours. By deploying specialized AI agents for competitive intelligence, financial analysis, market monitoring, scenario generation, and automated implementation, retailers could shift from reactive, error-prone workflows to proactive systems that respond to market dynamics. When learning capabilities are configured, the system evolves into a fully autonomous pricing engine that continuously improves its decision-making through feedback loops, historical performance analysis, and only exceptions are sent to human in the loop for critical decision making. This document also outlines current challenges, define quantifiable goals for the future state, and present a reference architecture for implementing agentic dynamic pricing using AWS services—including Amazon Bedrock, Amazon Bedrock AgentCore, leading Foundation Models, and optionally custom models via Amazon Nova Forge or Amazon SageMaker—to deliver the speed, accuracy, and scalability modern pricing demands.

Leveraging Agentic and Generative AI in the Retail Industry

The retail value chain presents significant opportunities for process improvement through Agentic and Generative AI-driven automation.



Beyond Traditional Automation

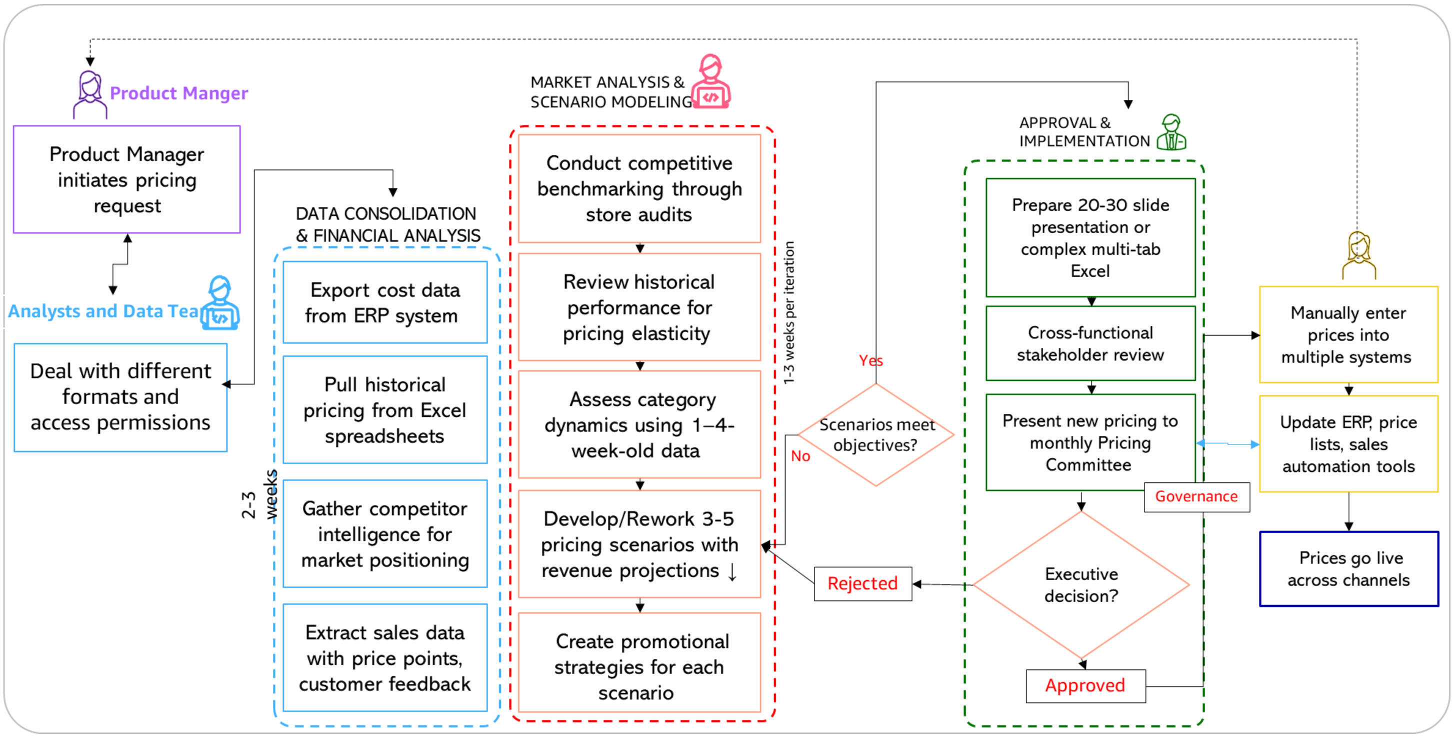
Previous automation technologies like rule-based systems and simple algorithms had significant limitations as they could only handle structured data and predefined scenarios. Agentic AI transforms pricing through:

- Dynamic processes are driven to achieve set goals, as against rule-based static processes.
- Processing both structured and unstructured data across multiple formats (sales data, market reports, social media sentiment)
- Dynamic and adaptive processes that evolve by learning from patterns and historic performances
- Handling market and competitive changes with agility and flexibility

Pricing Management: "As-Is" Process

The traditional pricing process represents a classic example of a spreadsheet and batch heavy manual, workflow that creates multiple pain points for both pricing teams and business stakeholders.

The sequential nature of the workflow, combined with heavy reliance on email communications, manual data gathering, and periodic reviews, results in slow response times and missed opportunities. Price changes can take weeks to implement, by which time market conditions may have already shifted.



Pricing Management - Current Business Process

Request Initiation & Data Collection: The product pricing process begins when a Product Manager initiates a pricing request, triggering manual data collection from several disparate sources such as the ERP system for cost data, Excel spreadsheets for historical pricing, competitor intelligence for market positioning, and sales information for past sales at various price points, and customer feedback. Analysts spend 1-2 weeks exporting data from disconnected systems, dealing with different formats and access permissions.

Data Consolidation & Financial Analysis: All disparate data flows into a consolidation phase where analysts manually merge the data either in Excel or ingest into data lakes — time-consuming activity with a high risk of inconsistencies. The consolidated dataset undergoes rigorous financial analysis—calculating cost of goods sold (COGS), sales volumes, margins, and profitability at various price points, and establishing price floors. These manual reconciliations represent major bottlenecks, taking 2-3 weeks and requiring multiple validation rounds with Finance teams.

Market Analysis & Scenario Modeling: Analysts examine competitive benchmarking through store audits, review historical performance for pricing elasticity patterns, and assess category dynamics—often working with data that is 1-4 weeks old. This feeds into scenario modeling where teams develop only 3-5 pricing scenarios with revenue projections and promotional strategies. A "Needs Rework" feedback loop frequently adds 1-3 weeks per iteration when scenarios fail to meet objectives.

Approval & Implementation: Cross-functional stakeholders review a comprehensive 20-30 slide presentation or complex multi tab excel before executive leadership makes final decisions at monthly Pricing Committee meetings. Upon approval, prices are manually uploaded (excel, into multiple systems (ERP, price lists, sales automation tools), followed by negotiations with key stakeholders that may trigger re-approvals. Finally, prices go live with ongoing performance monitoring through manual weekly reports and Excel-based variance analysis.

Key Pain Points

The current process suffers from critical inefficiencies spanning **6-10 weeks from implementation—** slow for dynamic market conditions. Manual processes pervade every stage with heavy reliance on Excel, manual data entry across systems, and high probability of errors. Siloed information forces analysts to reconcile inconsistent data formats rather than focus on strategic analysis. Limited analytical depth due to lack of data restricts teams to 3-5 scenarios when dozens might be relevant, while **data latency** of 1-4 weeks means decisions use outdated intelligence. **Communication bottlenecks** through email chains, sequential approvals with multiple loops, and meeting scheduling create significant delays. The process is fundamentally **reactive rather than proactive**, responding to market changes only after they occur, with **scalability constraints** preventing portfolio-wide reviews and **knowledge dependency** on individual analyst expertise creating organizational risk. The pricing process presents a great opportunity for transformation by leveraging Agentic AI.

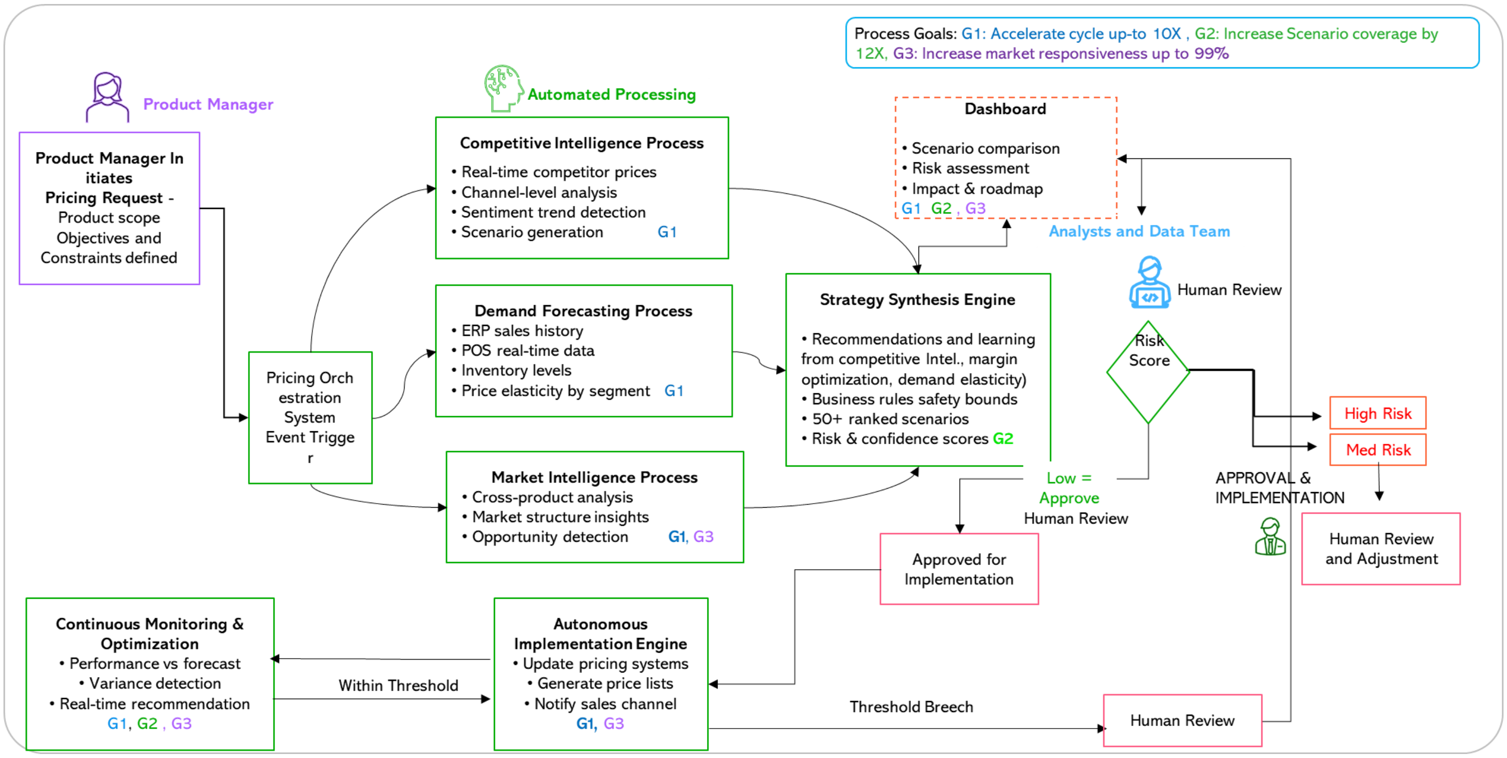
Pricing Management - Transformed (To-be) Business Process

The future of product pricing in retail companies lies in autonomous, intelligent systems that transform weeks of manual analysis into days of coordinated decision-making. By deploying specialized AI agents that work together under centralized orchestration, companies can shift

- **From reactive, resource-constrained pricing activities to proactive**, continuous optimization that responds to market dynamics, with exceptions only routed to humans. Over time, the AI will learn the patterns and can be configured to update the pricing dynamically in real time.
- From manual episodic tasks to specialized tasks across autonomous agents that operate continuously
- **From rule-based to intelligent process** that focus on achieving defined goals while handling data gathering, analysis, and scenario generation
- **From human-dependent to human-augmented**: freeing pricing teams to focus on strategic direction and stakeholder alignment

The To Be process is designed to achieve following goals for overall process and key tasks:

Metric	As-Is	To-Be	Improvement	Business Impact
Increase in Sceniro Coverage	3-5 Scenrios	50+ Data driven scenarios	12X-16X increase	✓ Top line and bottom line growth by exploring larger number of pricing scenarios ✓ Better Customer Experience
Pricing Cycle Time	6-10 weeks	2-4 Days	10X-35X Reduction	✓ Improved Operational Efficiency ✓ Better Customer Experience
Market Responsiveness	Weeks	Real-Time	99% increase	✓ Top line and bottom line growth by rapidly responding to market dynamics



Proposed Business Process for Pricing Optimization

The process begins when a Product Manager initiates a pricing request through a natural language chat interface, with conversation type inputs request to specify product scope, strategic objectives, and business constraints with guides and prompts guiding the manger to enter the right values. This single human action immediately triggers the orchestration system, which activates all downstream automated processes in parallel—eliminating the sequential bottlenecks of the as-is process where analysts spend 1-2 weeks manually collecting data from disparate sources.

Automated Parallel Data Collection & Analysis Four specialized automated processes execute simultaneously, each pursuing specific outcomes.

The **Competitive Intelligence Process** operates as an always-on monitoring system providing real-time competitive context—continuously tracking competitor pricing across all channels, analyzing consumer sentiment through social media and reviews, and identifying market trends through syndicated data feeds. When activated, it instantly provides current market positioning and generates dozens of scenario variations exploring different competitive strategies, channel-specific pricing approaches—replacing the 1–2-week manual process of gathering stale competitor intelligence.

The **Demand Forecasting Process** is designed to forecast demand at different price points and profitability for various conditions by continuously processing data from ERP systems (sales history, product data), inventory management systems (stock levels, turnover rates), point-of-sale systems (real-time transaction data) and calculates price elasticity across customer segments and channels - replacing weeks of manual analysis with automated pattern recognition.

The **Market Intelligence Process** performs comprehensive market analysis and competitive landscape assessment across multiple products simultaneously, identifying cross-product pricing opportunities that manual processes could miss.

The **Strategy Synthesis Process** combines recommendations from all pricing processes and generates several pricing scenarios applying parameters like competitive positioning, margin optimization, and demand elasticity, and importantly what agent has learnt from historical price recommendations and production managers' feedback. The agent further applies safety bounds based on defined business rules and strategic constraints. This automated process replaces the 1–3-week scenario modeling phase where analysts manually developed only 3-5 scenarios. The orchestration system synthesizes all inputs and can generate comprehensive pricing strategies with 50+ scenario options ranked by expected business impact, each accompanied by confidence scores, risk assessments, and transparent explanations detailing the reasoning behind recommendations. This replaces the 20-30 slide presentation preparation and multiple stakeholder review cycles.

Human Decision Point: Decision-makers receive interactive dashboards showing scenario comparisons, risk assessments, and implementation roadmaps—enabling approval within hours through streamlined digital workflows rather than waiting for periodic Pricing Committee meetings. The system presents clear decision points: scenarios meeting all business rules receive "Recommended" status, those with moderate risks are flagged "Review Required," and edge cases exceeding risk thresholds are routed to "**Human Exception Handling**" with detailed explanations of why human judgment is needed.

Exception Routing and Human-in-the-Loop: When the system encounters scenarios outside predefined confidence thresholds—such as unprecedented competitive moves, extreme demand volatility, or margin compression beyond acceptable limits —it automatically routes for decision to human experts with comprehensive context: the specific rule violated, historical precedents, risk quantification, and alternative options. This ensures human oversight for high-stakes decisions while automating routine pricing updates, achieving the process-level goal of balancing speed with safety.

Autonomous Implementation & Continuous Optimization Upon approval, the system automatically coordinates execution across all systems in parallel—updating pricing databases, generating customer-specific price lists and triggering sales notifications—replacing the manual 2–4-week implementation phase where prices are uploaded or ingested into multiple systems manually. The system maintains continuous performance monitoring, autonomously tracking actual results against projections, identifying variances, and recommending real-time adjustments when market conditions shift. When performance deviates beyond defined thresholds, the system routes recommendations back to human decision-makers, creating a closed-loop system where pricing decisions are continuously refined based on market feedback while maintaining human control over strategic changes.

Key Benefits

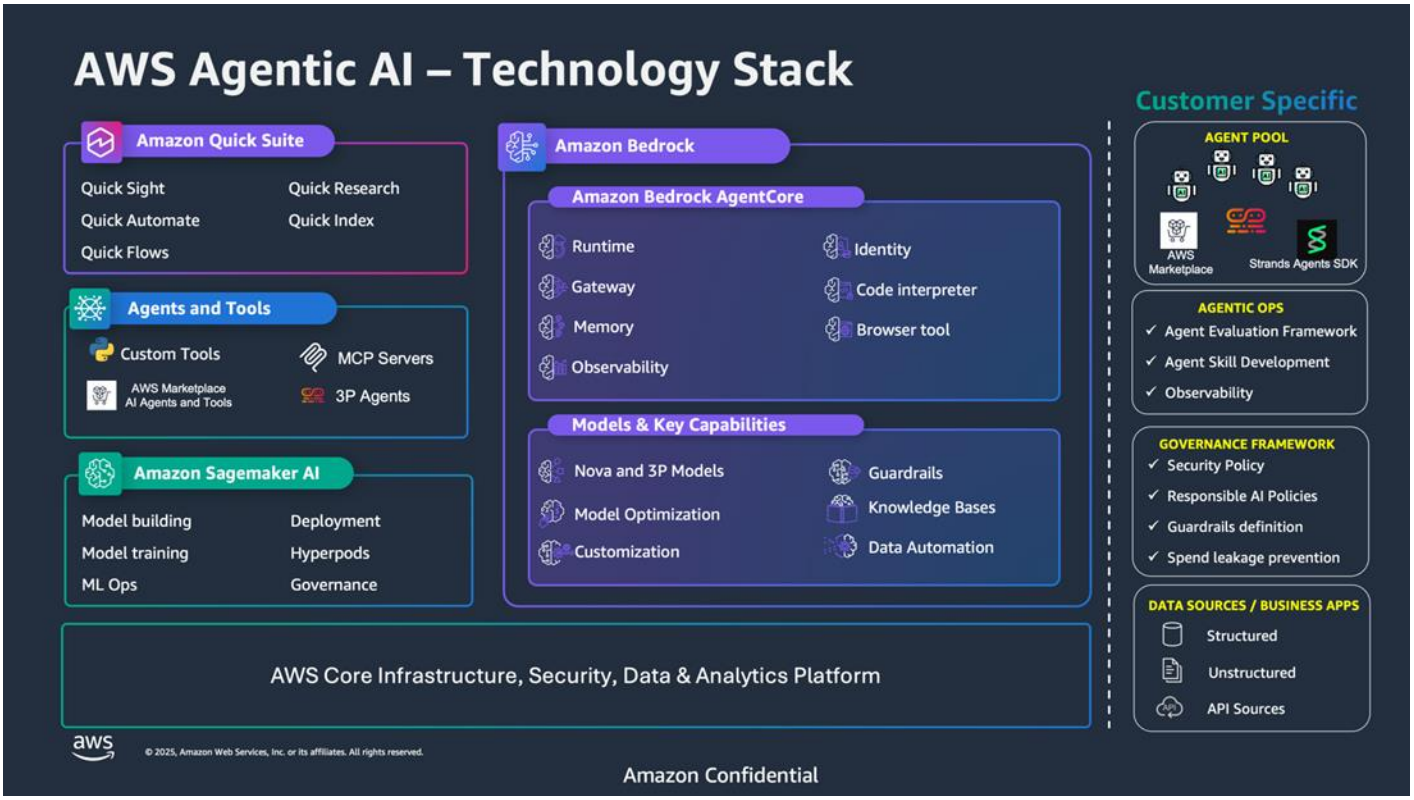
This agentic architecture delivers **85-90% time reduction** (from 6-10 weeks to 2-4 days), enabling companies to respond to competitive moves and market shifts with unprecedented speed. The system achieves a **5-15% revenue increase** through optimized pricing while reducing manual effort by 80%, freeing pricing teams to focus on strategic initiatives rather than data wrangling. Also, it scales effortlessly across entire product portfolios, maintains consistent methodology, and operates proactively anticipating market shifts rather than reacting to them. Companies implementing agentic pricing systems report 99% faster response times to market changes, 95% recommendation accuracy, and the ability to explore 10-20x more pricing scenarios than traditional approaches, leading to better-informed decisions and superior business outcomes.

Guidance for To-Be Architecture and Implementation

This section provides detailed guidance for implementing the AI-powered pricing system, including architectural decisions, technology choices, and implementation of best practices.

AWS Agentic AI Technology Stack

The solution leverages AWS's comprehensive AI and cloud services to deliver a scalable, secure, and performant pricing system.

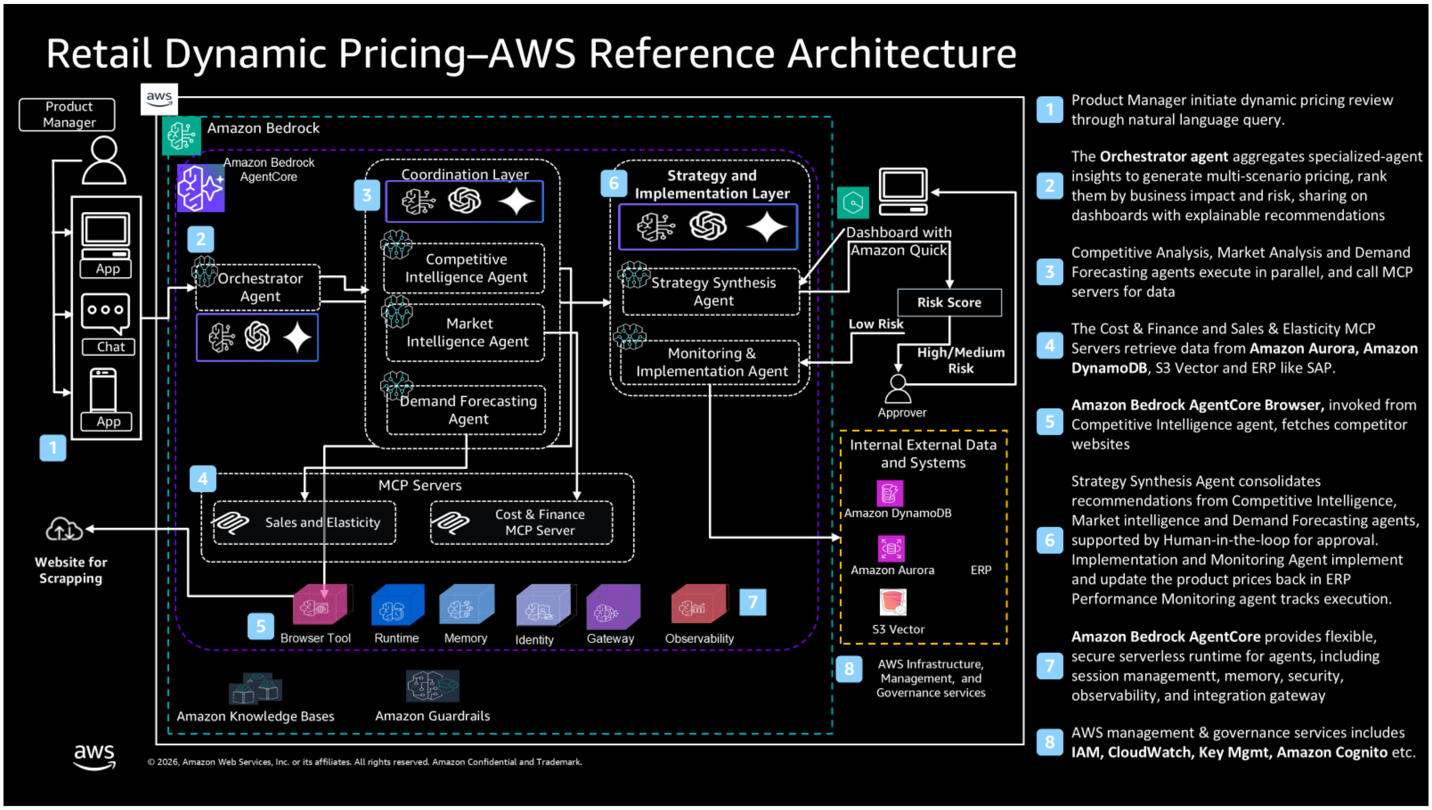


AWS’ Agentic AI stack consists of three key components:

- Agent-powered applications, like Amazon Quick Suite and Amazon Q for Business, provide fully managed software/applications for business users and developers.
- Tools for building AI agents, which include custom tools from AWS, MCP servers, and third-party tools.
- Agentic AI foundation, which includes Amazon Bedrock, Amazon Bedrock AgentCore, Amazon Bedrock Guardrails, and Amazon SageMaker AI, for building, training, and deploying enterprise-grade AI agents

Refer to Appendix A for overview of these components.

Agentic Solution for Retail Dynamic Pricing - Reference Architecture



The core building blocks of the Reference Architecture are:

Amazon Bedrock

Transforming retail pricing requires intelligent systems that can analyze vast amounts of market data, interpret complex competitive dynamics, and make informed pricing decisions adhering to business objectives and constraints. **Amazon Bedrock** enables this through secure access to leading foundation models and enterprise-grade capabilities including knowledge bases and guardrails for compliant retail operations. Refer appendix for overview Amazon Bedrock.

Foundation Models for Dynamic Pricing

Retail pricing requires AI models that can efficiently analyze market trends, interpret competitive positioning, assess demand elasticity, and make accurate pricing recommendations. **Amazon Nova Understanding Models** offer optimal and cost-effective options for market data analysis and competitive intelligence gathering, while **Anthropic Claude Sonnet** can be leveraged for complex reasoning tasks like multi-factor pricing synthesis and revenue optimization, and **Anthropic Claude Opus** manages the most demanding multi-step analytical processes requiring maximum capability such as pricing optimization. Organizations can choose from over 80+ models from leading AI providers and hundreds of specialized models from **Amazon Bedrock Marketplace**. Furthermore, large retailers with highly specialized pricing rules and strategies would benefit from distilling **Amazon Nova Premier** into smaller, efficient student models and further customizing them through **Amazon Nova Forge** using their proprietary pricing data to create purpose-built pricing models that balance domain expertise with general intelligence.

Amazon Bedrock Data Automation can be used to extract pricing-relevant information through fully managed processing of market reports, customer transaction data, and inventory feeds. For real-time competitor price monitoring from websites, AgentCore Browser with Nova Act provides the web scraping capability, with the extracted data then fed into the pricing pipeline.

Reliable & Robust Pricing Process with Strong Governance and Quality Assurance:

To ensure prices are set in adherence to organizational pricing policies, guidelines **Amazon Bedrock Guardrails** can be used. Content filters and sensitive information filters can be configured to block prompts and responses that explicitly seek or describe anti-competitive or unfair strategies (for example, “set prices below cost to drive out competitor X” or “offer a special price only for this named customer”), and to restrict the model’s use of customer- or competitor-identifying data. These safeguards reduce the risk that the model will assist in designing predatory or discriminatory pricing strategies, while the actual enforcement of pricing floors and non-discrimination rules remains in application-level logic.

Amazon Bedrock AgentCore

Deploying production-ready pricing agents requires scalable infrastructure that enables secure orchestration of multi-agent workflows and integrates with existing retail systems. **Amazon Bedrock AgentCore** enables this through serverless runtime with session isolation, memory retention across pricing cycles, and seamless integration with enterprise systems such as inventory management, point-of-sale systems, e-commerce platforms, and competitive intelligence APIs. Further, Amazon Bedrock AgentCore supports leading Agentic frameworks such as Strands Agents, , LangGraph, CrewAI, LlamaIndex, OpenAI Agents SDK, Google ADK, and Microsoft AutoGen. This flexibility allows customers to use their existing investments and skills while building and deploying secure, enterprise-grade AI agents on AWS.

Executing End-to-End Pricing Process Leveraging Purpose-Built Agents:

The retail dynamic pricing solution employs a multi-agent orchestrator pattern. Product Manager initiates dynamic pricing review through natural language query defining objectives and product line or boarder categories (e.g. milk or even dairy products. The Orchestrator agent works in parallel with the Competitive Intelligence, Market Intelligence, and Demand Forecasting agents to drive pricing decisions. **Amazon Bedrock AgentCore Browser** is the managed Chrome browser service that lets agents interact with web pages for web scrapping. It supports both Strands/Playwright and Nova Act as automation drivers and fetch the price for the products from the external competitor websites. The two MCP Server —**Cost & Finance and Sales & Elasticity**—fetch data from Aurora, S3, DynamoDB and ERP systems like SAP. The **Strategy Synthesis Agent** combines recommendations from Competitive Intelligence, Margin Intelligence, and Demand Forecasting agents, with Human-in-the-loop approval for workflows. Implementation and Monitoring Agent execute price updates across ERP and sales systems with low-risk changes are auto-implemented; medium and high-risk changes require human approval before execution, tracks actual results against projections and recommends adjustments when deviations exceed thresholds, routing significant variances to human decision-makers for review — creating a closed-loop feedback system. Each product pricing decision must be processed independently to prevent data leakage between SKUs and categories. **Amazon Bedrock AgentCore Runtime** organizes sessions at the pricing-group level — product families, sub-categories, or entire categories — rather than processing individual SKUs in isolation. This means related products, such as different pack sizes of the same item, are evaluated together within a single session, ensuring pricing coherence across the product line and reflecting how retailers actually make coordinated pricing decisions Routine pricing adjustments are handled through deterministic rules for cost efficiency and speed, while the multi-agent system is reserved for complex pricing scenarios requiring competitive analysis, elasticity modeling, and cross-category impact assessment. This tiered approach supports the scale of large retailers (hundreds of thousands of SKUs) without incurring prohibitive LLM costs. While full automation (straight-through processing) is the objective, human validation checkpoints are maintained for high-value items, significant price adjustments, or decisions that cross predefined business thresholds

Secure Access to Enterprise Systems:

Systems such as inventory management, databases, and ERP platforms, can be achieved using **Amazon Bedrock AgentCore Identity** that allows granting appropriate credentials to the agents. When processing pricing decisions requires coordination across multiple specialized agents—such as for Demand Forecasting, Competitive Intelligence, Market Intelligence —Agent-to-Agent (A2A) protocol of Amazon Bedrock AgentCore Runtime can be used to enable secure inter-agent communication through open standards that support external agent integration through open standards that support external agent integration.

Integrating Pricing Agents with Enterprises and External Systems:

While implementing agents for dynamic pricing, it is critical that agents can integrate with other enterprise and third-party systems like inventory management systems (e.g., SAP, Oracle), e-commerce platforms (e.g., Shopify, Magento), point-of-sale systems and competitive intelligence platforms. Agents enable this through unified **MCP**-compatible endpoints that transform legacy APIs into agent-ready tools using **OpenAPI** specifications and **AWS Lambda** functions without custom integration code. **Amazon Bedrock AgentCore Gateway** provides authentication using AWS IAM, three-legged OAuth (3LO) for user-delegated access to third-party systems, two-legged OAuth (2LO) for machine-to-machine authentication, and secure credential exchange with automatic token refresh.

Amazon Bedrock AgentCore Observability provides end-to-end tracing, capturing key performance data for the end-to-end dynamic pricing workflow with detailed audit trails, while real-time **Amazon CloudWatch** dashboards track critical metrics including processing latency, token consumption, session counts, and error rates. **OpenTelemetry**-compatible telemetry emission enables seamless integration with existing enterprise monitoring tools like Datadog and Dynatrace, and rich metadata tagging with contextual filtering simplifies issue investigation and quality maintenance across high-volume dynamic pricing operations.

Amazon Quick Suite Dashboard

Pricing managers and merchandising teams require intuitive access to AI-powered pricing capabilities without technical expertise. **Amazon Quick Suite** provides business users with a natural language interface, enabling pricing managers to conduct pricing investigations through **Amazon Quick Research**, analyzing "Why did price increase for Product X?" or "What's driving competitor pricing in the Electronics category?" , create custom pricing workflows through **Amazon Quick Automate** building approval processes and exception handling without coding, visualize pricing metrics and market trends through **Amazon Quick Sight** dashboards and, access consolidated pricing policies and market intelligence through **Amazon Quick Index**——all while maintaining enterprise security and governance controls.

Agent Evolution through continuous Learning

AI agents built on the AWS technology stack continuously improve their intelligence by remembering key facts, learning from past experiences, incorporating closed-loop feedback, and adapting to organizational patterns over time working backwards from defined goals to drive compounding performance gains. **Amazon Bedrock AgentCore Memory** retains pricing context across interactions at multiple levels through two complementary memory types that work together to make agents more accurate and contextually relevant with each completed pricing cycle. Short-term memory holds the active pricing analysis data — current demand signals, competitor prices, and in-progress calculations — for the duration of a pricing session, configurable up to a set of days. Short-term memory serves a dual purpose: it keeps agents coherent throughout a pricing session without requiring analysts to re-establish context at each interaction, and it preserves complete decision context for pricing manager reviews — providing full traceability of agent reasoning without manual reconstruction. Long-term memory persists pricing strategies, historical performance patterns, and learned elasticity models that remain available indefinitely across all pricing cycles. Within long-term memory, two built-in strategies are particularly relevant to pricing- Semantic memory enables agents to remember key facts and aggregates knowledge about entities such as customers, products, SKU prices, and market conditions, giving agents a persistent understanding of the pricing landscape. Episodic memory stores structured records of past pricing episodes — including input conditions, reasoning traces, actions taken, and outcomes — enabling agents to learn from previous decisions. Rather than storing every raw interaction, AgentCore Memory identifies meaningful moments from each pricing episode — such as a competitive price match that drove a conversion increase, an elasticity estimate that accurately predicted demand response, or a markdown strategy that cleared excess inventory without margin erosion. These moments are summarized with relevant context into compact records the agent can retrieve and apply to future pricing cycles. Episodic memory operates through a three-step strategy: extraction analyzes in-progress episodes and identifies meaningful pricing decisions and their context; consolidation combines extractions into a single structured episode record upon session completion; and reflection uses foundation models to identify patterns across similar episodes — such as which pricing strategies succeeded in specific competitive conditions — creating durable knowledge that improves future recommendations. Through reflection, agents move beyond recalling what happened in past pricing cycles to understanding why certain outcomes occurred — which competitive responses triggered price wars, which demand signals were leading indicators, and which elasticity assumptions held under different market conditions. This turns accumulated pricing experience into actionable guidance that compounds over time. Prices are adjusted based on demand, competitor feeds, and stock levels. Learning occurs by observing the revenue, conversion, and margin impacts of past pricing decisions, and updating elasticity estimates accordingly. An agent that has processed several hundred pricing cycles increasingly becomes more accurate at determining which pricing strategies succeed in specific competitive environments, which demand patterns indicate upcoming elasticity shifts, and which inventory signals warrant aggressive markdowns versus hold strategies — enabling progressively better pricing recommendations as the agent's accumulated knowledge grows.

Addressing Compliance, Security and Privacy considerations

The proposed architecture delivers comprehensive security, compliance, and privacy controls to meet enterprise requirements.

Regulatory Compliance and Governance

Amazon Bedrock AgentCore Observability provides step-by-step visibility into agent decision-making processes with detailed trace data, enabling root cause analysis for pricing disputes, and regulatory audits. This transparency is essential for demonstrating compliance with few country specific acts like, Robinson-Patman Act preventing price discrimination between different purchaser, Federal Trade Commission (FTC) regulations ensuring fair and transparent pricing practices, State consumer protection laws meeting jurisdiction-specific pricing requirements, international pricing regulations Compliance with EU consumer protection directives and similar global standards. **Amazon Bedrock** maintains comprehensive compliance certifications, with SOC 2 and ISO 27001 being particularly relevant to retail pricing operations — ensuring security, confidentiality, and processing integrity for proprietary pricing strategies and competitive intelligence data. PCI DSS compliance provides additional assurance where pricing systems interact with transaction or payment-adjacent data. The platform also holds SOC 1, SOC 3, CSA STAR Level 2, and FedRAMP High certifications, with regular third-party audits validating security controls. Retailers can leverage **AWS Artifact** to access compliance reports and certifications for regulatory submissions, vendor risk assessments, and customer due diligence requirements. Complete audit trails via **Amazon CloudWatch** and **Amazon CloudTrail** capture all pricing decisions, model invocations, configuration changes, and access patterns, creating immutable records for regulatory examination.

Responsible AI and Bias Mitigation

Responsible AI principles require retailers to evaluate AI systems for fairness across diverse customer groups and maintain transparency in pricing decision-making. **Amazon Bedrock** provides automated model evaluation capabilities using curated datasets to assess bias across demographic factors, with support for custom evaluation datasets reflecting retail-specific considerations such as Geographic bias ensuring pricing doesn't unfairly disadvantage customers in specific regions beyond legitimate cost differences; Temporal bias validating that time-based pricing (peak vs. off-peak) doesn't create discriminatory patterns; Purchase history bias ensuring personalized pricing based on customer behavior doesn't create unfair advantages or disadvantages; Device/channel bias preventing price discrimination based on whether customers shop via mobile, desktop, or in-store. **AWS AI Service Cards** provide comprehensive documentation of each foundation model's intended use cases, limitations, responsible AI design principles, and performance characteristics, enabling retailers to make informed model selection decisions. Model evaluation jobs generate detailed fairness of metrics, accuracy measurements, and robustness assessments suitable for internal governance and regulatory submission. **Amazon Bedrock Guardrails** enforce content policies to prevent biased or inappropriate pricing recommendations.

Data Protection and Privacy

Retail pricing data contains sensitive customer information including purchase history, preferences, and potentially personally identifiable information (PII). The system implements stringent protection controls to safeguard this data. **Amazon Bedrock** ensures data remain under customer control through encryption in transit and at rest (e.g. Pricing history and analytics encrypted), with no storage or logging of prompts and completions by AWS (e.g., Pricing queries and recommendations). Data used for model customization creates private copies ensuring information is never shared with model providers or used to improve base models. **AWS PrivateLink** enables private connectivity from VPCs to **Amazon Bedrock** and **Amazon Bedrock AgentCore** without internet exposure, supporting GDPR compliance. **Amazon Bedrock AgentCore Runtime** enforce session isolation where each dynamic pricing runs in a dedicated microVM, and after session termination, the entire microVM is terminated and memory is sanitized, ensuring no residual price data persists. **Amazon Bedrock Guardrails** provide additional PII protection through detection to prevent inadvertent disclosure in agent responses.

Conclusion

The Retail Dynamic Pricing Agent represents a fundamental shift in how companies approach pricing strategy. By breaking complex pricing decisions into specialized, coordinated components, organizations are cutting pricing cycle times by 85-90%—what used to take 6-10 weeks now happens in 2-4 days. This speed advantage means companies can respond to competitor's moves and market changes while opportunities are still fresh.

The business impact is substantial and measurable. Companies using this approach are seeing **5-15% revenue growth** through smarter pricing decisions, while their pricing teams spend 80% less time on manual data work. That freed-up capacity goes straight into strategic thinking and market analysis. Response times to market shifts have improved by 99%, and pricing recommendations hit the mark of 95% of the time.

Perhaps most importantly, teams can now evaluate 10-20 times more pricing scenarios than before. This isn't just about speed—it's about exploring possibilities that were previously impractical, leading to better-informed decisions and stronger business results. The system scales naturally across entire product catalogs while keeping methodology consistent, and it anticipates market movements rather than simply reacting to them.

The modular design means each component can be updated, scaled, or enhanced independently as business needs to change. Each specialized element handles its specific aspect of pricing—competitive intelligence, demand forecasting, inventory optimization—working together to deliver recommendations that adapt real-time to market conditions.

This isn't about replacing human judgments with AI. It's about giving pricing teams better tools and more time to focus on what matters: strategic decisions that drive competitive advantage. The system augments human expertise with speed, consistency, and analytical depth that transforms pricing from a periodic exercise into a **continuous competitive advantage.**

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Appendix A: AWS Agentic AI Tech Stack

Amazon Bedrock

Amazon Bedrock is a fully managed generative AI service that provides secure access to high-performing foundation models from leading AI companies through a unified API, enabling organizations to build and scale AI applications without managing underlying infrastructure. It offers comprehensive capabilities including model customization, knowledge bases, and built-in guardrails to ensure responsible AI deployment while maintaining data privacy and regulatory compliance.

Amazon Bedrock AgentCore

Amazon Bedrock AgentCore is a serverless platform for building, deploying, and running AI agents at scale. Key capabilities include:

- AgentCore Runtime — serverless agent deployment with automatic scaling and pay-per-use pricing, requiring no infrastructure management
- AgentCore Memory — persistent agent state with both semantic and event-based memory for maintaining context across pricing sessions
- AgentCore Identity — centralized credential management (OAuth 2.0, API keys, client certificates) for secure agent access to enterprise systems
- AgentCore Gateway — exposes existing APIs as agent-callable tools via MCP protocol, enabling integration with pricing, inventory, and market data services
- AgentCore Browser — cloud-based browser tool for real-time web interaction, supporting competitive intelligence gathering
- AgentCore Code Interpreter — secure sandboxed execution environment for dynamic computations such as elasticity modeling and pricing simulations
- AgentCore Observability — real-time monitoring, tracing, and session-level visibility into agent execution and performance

Agentic AI Memory Architecture

AgentCore Memory supports both short-term and long-term memory for pricing agents:

- Short-term memory: Conversational events within a session, providing immediate context for the current pricing analysis
- Long-term memory: Persistent knowledge extracted from events across sessions, implemented through configurable memory strategies:
    - Semantic memory: Extracts and consolidates factual knowledge — including aggregated information about entities such as customers, products, and SKU pricing patterns — enabling agents to recall what they know
    - Episodic memory: Structured records of past pricing episodes, including input conditions, reasoning traces, actions taken, and outcomes, with reflections that enable agents to learn from their own experience

Long-term and episodic memories are often backed by vector databases, relational stores, or hybrid solutions. Crucially, episodic memory enables *experience-based learning*: the agent not only recalls facts but also how specific strategies are performed in real contexts.

Amazon QuickSight

Amazon QuickSight provides business intelligence and visualization capabilities:
• Real-time dashboards for pricing performance
• Interactive visualizations of market trends
• Embedded analytics in applications
• ML-powered insights and anomaly detection
• Secure data access and sharing

Appendix B: Addressing Compliance, Security, and Privacy Needs

These are key concerns for **any company implementing Agentic and Generative AI projects.**

Regulatory Compliance

- SOC 2 Type II - security, availability, and processing integrity controls, directly applicable to systems handling pricing decisions and customer data.
- ISO 27001 - Information security management
- GDPR - European data protection
- State and federal price gouging laws that restrict excessive price increases during declared emergencies.
- FTC Act Section 5, which prohibits unfair or deceptive pricing practices — including algorithmically generated prices that could mislead consumers.
- EU Omnibus Directive (2019/2161), requiring disclosure of the lowest prior 30-day price alongside any advertised reduction.
- Emerging algorithmic pricing transparency obligations under the EU AI Act.

Data Encryption

All data within the system is encrypted both at rest and in transit. Data at rest — stored in DynamoDB, S3, and CloudWatch Logs — is encrypted using AES-256 via AWS Key Management Service (KMS), with automated key rotation to reduce the risk of key compromise. Data in transit between services, agents, and API consumers is secured using TLS 1.3. SSL/TLS certificates are managed through AWS Certificate Manager (ACM), eliminating manual certificate provisioning and renewal

Access Control

Access control is enforced through AWS IAM, following a defense-in-depth approach. Role-based access control (RBAC) assigns permissions based on function — e.g., product managers, operations staff, and administrators each receive only the access their role requires, in line with the least privilege principle. Human users authenticate with multi-factor authentication (MFA), while service-to-service communication between agents, Lambda functions, and AWS resources (Bedrock, DynamoDB, S3) is authenticated via IAM roles — no hardcoded credentials. All access events are logged through AWS CloudTrail for audit and compliance purposes.